Electric Vehicle Routing Problem With Capacitated Charging Stations and Dynamic Electricity Prices

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***Abstract*—With rapid electrification of transportation systems in recent years, the electric vehicle routing problem (EVRP) has received increasing attention. In order to serve a set of valuable transportation tasks using a fleet of electric vehicles (EVs), EVRP intends to jointly determine their routes and the charging schedules, with the aim of optimizing service reward and charging cost under state of charge (SOC) constraints of battery. EVRP is a difficult nonlinear integer program, and due to its intrinsic NP-hardness early EVRP studies often oversimplify the charging process and assume unlimited capacity of charging stations (CSs) and constant electricity price. The contribution of this article lies in twofold. First, to better characterize the practical charging process, this article introduces a new graph model that can take the CS capacity and dynamic electricity price into full consideration. Second, we design a series of transformation techniques, including reformulation SOC dynamics, Lagrangian relaxation, exact linear programming relaxation, etc, which eventually convert the original EVRP into an approximate linear program. Extensive simulations across 14 instances demonstrate that, the proposed approach is capable of solving large-scale instances over 500 nodes, and is 100 times faster than traditional analytic methods while achieving very close optimality. While comparing with the popular heuristic method adaptive large neighborhood search (ALNS), our method is several times faster and obtains much better optimality.**

***Index Terms*—Electric vehicle (EV), integer programming, linear programming (LP), routing problem.**

I. Introduction

ELECTRIC vehicles (EVs) are becoming increasingly popular due to their environmentally friendliness and sustainable nature. In recent years, a large amount of EVs have been put into use in public transportation sectors, such as buses, logistic vehicles, taxis, and postal vehicles. In such sectors, the operator often needs to manage a fleet of EVs and determine the routes for them to complete a set of geographically dispersed transportation tasks. Meanwhile, the operator also needs to determine where and when to charge the battery during the trips due to EVs' limited capacity. Mathematically, such an operation task can be formulated as an optimization problem, i.e., the electric vehicle routing problem (EVRP), which aims at optimizing the service reward and charging cost while satisfying the state of charge (SOC) constraints of EVs, by properly selecting the transportation routes and scheduling the charging. Indeed, EVRP is an important variant of the vehicle routing problem (VRP), which is one of the most widely studied problems in the field of operations research.

A. Related Work

In recent years, a lot of researchers have been engaged in studying the EVRP, which, however, is a nonlinear integer program (NIP) and NP-hard in the strong sense [1], [2], [3], [4], thus poses a huge challenge for the design of computation-efficient algorithms. In literature, the EVRP can be solved using either exact methods [5], [6], [7] or heuristic approaches [8], [9], [10]. Exact methods are designed to find optimal solutions to the EVRP, but their computational efficiency is often limited by the problem's complexity. Hiermann et al. [5] employed the branch-and-price solver for small-scale instances. Compared to the traditional branch-and-bound method, which failed to solve 10 or 15-customer cases within 2 h, branch-and-price successfully solved all 15-customer cases, demonstrating its effectiveness for small-scale problems. However, its computational efficiency remains a limitation, as some cases took over 900 s, which is relatively long for such small instances. To address the limitations of small-scale solvers, Ceselli et al. [6] proposed a branch-and-cut-and-price (BCP) algorithm. The key strength of BCP lies in its decomposition strategy, which breaks each feasible route into smaller paths and computes them independently for each pair of charging stations (CSs), enabling parallel processing. This approach significantly improves scalability, allowing the algorithm to solve instances with up to 30 customers. However, BCP still struggles with larger and more complex scenarios due to its high computational complexity. Building upon this foundation, Lam et al. [7] proposed a hybrid BCP algorithm that integrates integer programming via Dantzig–Wolfe decomposition for routing and constraint programming for scheduling, utilizing logic-based Benders decomposition. Their method introduces strengthened Benders cuts and performs polyhedral analysis to enhance computational efficiency. This enhancement enables the algorithm to

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tackle 34% instances with up to 100 customers, showing its capability to handle larger customer volumes. However, even with these improvements, exact algorithms face significant challenges in solving large-scale problems within a reasonable timeframe.

In contrast to exact approaches, heuristic methods offer significant reductions in computation time. These approaches prioritize computational efficiency over optimality, leading to faster solutions for EVRP [9]. Schneider et al. [11] introduced a hybrid heuristic that combines a variable neighborhood search algorithm with a tabu search heuristic to solve EVRP. Although this algorithm is several times faster than CPLEX, its evaluation was limited to small-scale instances with up to 15 customers, indicating that VNS and TS are more suitable for medium-sized problems. Keskin and Çatay [12] further developed an adaptive large neighborhood search (ALNS) algorithm, utilizing removal and insertion mechanisms to dynamically adjust solutions. In each iteration, customers and CSs are removed and then reinserted as needed. While ALNS demonstrates fast performance in solving large-scaled EVRP instances, it tends to fall into local optima, which may affect its effectiveness in certain scenarios. To address this limitation and enhance global search capabilities, Duan et al. [10] incorporated the simulated annealing method into the ALNS framework. This approach allows the algorithm to accept suboptimal solutions with a certain probability, thereby increasing solution diversity and improving the ability to explore the global solution space. However, it is worth noting that heuristic algorithms typically lack a guarantee of solution optimality.

However, there are still research gaps for EVRP. Clearly, the computational performance in terms of both speed and optimality of current methods is still not satisfactory, especially for large-scale problems. Therefore, the first motivation of this research is to develop new method that is able to achieve near-optimal solutions with fast speed. The second motivation is that previous studies often oversimplify the charging process and do not consider limited charging capacities at CSs, as well as dynamic charging price that is a major feature of smart grid [13], [14], [15], [16]. Indeed, these factors are realistic and very important in the operation of EVs, but they certainly introduce extra complexity to the design of computation-efficient algorithms. Therefore, most existing models neglect these factors and simply assume that CSs are always available to EVs upon arrival [17], [18], [19], [20], [21] and the charging price is a constant. However, this assumption is not true for public CSs, which typically have a fixed number of charging piles that can only serve a limited number of EVs, and the charging price fluctuates depending on time and location.

B. Contributions

This article aims to bridge such research gaps. That is, we not only take the capacity of CSs and dynamic charging prices into full consideration with a novel model, but also devise a very efficient method to solve the problem in polynomial time with near-optimal results. The main contributions of this article are twofold.

TABLE I
KEY NOTATIONS

Given Parameters	
ϵ	Energy consumption rate of EVs
$\mathcal{C}$	Set of charging stations
$\mathcal{K}$	Set of EVs
$\mathcal{R}$	Set of customer nodes
d_{ij}	Travel distance defined by Euclidean distance from node i to j
E_0	Initial battery energy of EVs
E_{max}	Battery capacity
g_i	Amount of charge during one time slot at the charging node
m_i	Revenue of serving the node i
p_{ij}	Energy consumption traveling from node i to node j
q	Velocity of EVs
t_i	Visiting time of customer requests at node i
w_i	Cost of charging on one charging node i
Decision Variables	
x_{ij}^k	Binary variable indicating whether EV k traverses arc (i,j)
Other Notations	
E_i^k	State of charge of EV k at node i
f_i^k	Binary indicator representing whether EV k visits node i
$G(\mathcal{V},\mathcal{A})$	Directed graph, where $\mathcal{V}=\{v_0,\cdots,v_n\}$ consists of the start depot, end depot, customer nodes, and charging nodes. The set of arcs between nodes are denoted by $\mathcal{A}=\{a_{12},\cdots,a_{ij}\}$

1) In order to characterize the limited number of charging piles of CSs and dynamic charging prices, we propose a novel graph model, which regards the charging actions as nodes, and dynamic prices as their costs, respectively. By this means, we can uniformly model both transportation tasks and charging actions as nodes with corresponding revenues or costs.
2) In order to enable efficient computation, we devise an integrated method including several techniques, such as reformulation SOC dynamics, approximate Lagrangian relaxation, exact linear programming (LP) relaxation, which can eventually convert the original EVRP in the form of NIP into an LP, so that the problem can be efficiently solved in polynomial time with near-optimal results.

This article is organized as follows. Section II describes the system and introduces a new graph model to incorporates the capacitated CSs and dynamic charging prices. Then, the objective function and constraints of EVRP are formulated on the graph model in Section III. A novel integrated method is proposed in Section IV, whose performance is evaluated by numerical simulations in Section V. Finally, this article is concluded in Section VI.

II. System Description and Graph Model

In this article, we will use a transportation network to model EVRP and its main components are stated below. All key notations are summarized in Table I.

A. Electric Vehicles and Customer Nodes

As shown in Fig. 1, we consider a set of homogeneous EVs denoted as $\mathcal{K}$, to serve a set of valuable customer requests and charge at CSs from time to time. Each EV $k \in \mathcal{K}$ can be characterized by a tuple $\{E_0, E_{\max}, q, \epsilon\}$, where E_0 is seen as the initial energy when leaving the start depot, $E_{\max}$ as the

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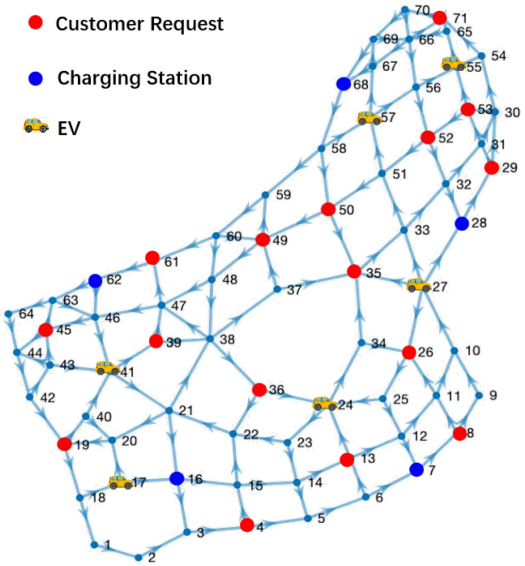


Fig. 1. Scenario of EVRP.

battery capacity, q as the velocity of EV and ϵ as the energy consumption rate.

Let $\mathcal{R}$ denotes the set of customer requests, or, customer nodes. Before the fleet of EVs begins service, customers will broadcast their requests. For a customer node $i \in \mathcal{R}$, the tuple $\{t_i, s_i, m_i\}$ presents a request including the visiting time t_i, service duration s_i, and revenue m_i of serving this request. In this article, we assume that every node i only requires at most one EV to serve, and EVs should arrive at the customer node before its visiting time t_i. If a person makes multiple requests at the same time, then multiple customer nodes will be generated.

B. Charging Nodes

Due to limited battery capacity, EVs need to charge at charging pile in a CS from time to time during the course of service. In this article, we consider charging actions as nodes in the transportation network. That is, a charging node $j \in \mathcal{C}$ with tuple $\{t_j, \Delta t, w_j\}$ denotes a charging action at time slot t_j with period Δt and charging prices w_j. It is important to notice that for two different charging nodes $j \neq j' \quad j, j' \in \mathcal{C}$, there are three possibilities.

1) Two charging actions happen at the same time, but on different charging piles.
2) Two charging actions happen on the same charging pile, but at different time slots.
3) Two charging actions happen on different charging piles and at different time slots.

If an EV charges on the same pile for several periods, it can be modeled as sequentially visiting a series of charging nodes. It is also notable that a charging node can only be visited by one EV at most. On the other hand, a CS usually has multiple charging piles that correspond to multiple charging nodes during the same period, thus our model allows multiple EVs to charge simultaneously in a CS.

It is also worth noting that the charging price w_j is very general, and it can be dynamic and different from each other. In such a way, we can capture the variation of electricity prices at different locations and different time slots. Therefore, with our model and solution approach, which will be presented later, the grid operator can use dynamic electricity prices as demand response signals to regulate the charging behavior of EVs.

C. Transportation Network

With customer nodes and charging nodes, the transportation network is then modeled as a directed graph. Let $\mathcal{V}$ denotes the set of nodes in the network, including start depot v_0, end depot v_n, the customer nodes $\mathcal{C}$, and charging nodes $\mathcal{C}$

$$\mathcal{V} = \mathcal{R} \cup \mathcal{C} \cup \{v_0, v_n\}. \tag{1}$$

It is important to notice that we can always arrange the index of nodes in the order of visiting time, i.e., $i < j$ if $t_i \leq t_j$ for any $i, j \in \mathcal{V}$. As such, a directed arc a_{ij} from node i to node j is valid only if

$$t_i + s_i + \frac{d_{ij}}{q} \leq t_j \tag{2}$$

where d_{ij} is the distance between two nodes. The above formula means the visiting time of node j must be later than the visiting time of node i plus the service time and the driving time. If i is a charging node, $s_i = \Delta t$; if both i and j are charging nodes on the same charging pile, then $d_{ij} = 0$.

By using (2), a large number of time-invalid arcs will be removed. Without loss of any generality, we can only consider the set of valid arcs $\mathcal{A}$ that satisfy (2). Then, the graph of the transportation network is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{A})$, on which our later analysis will proceed.

Compared with previous studies of EVRP, where both the varying electricity price and the CS capacity constraint have been rarely considered, our new model of charging actions as charging nodes enjoys three immediate advantages.

1) We can use a unified graph model to represent both the customer requests and charging actions for EVRP so that the same set of mathematical tools can be devised to handle both routing and charging decisions.
2) It is ready to introduce w_j to characterize the time-varying and spatially different electricity prices so that the charging cost can be calculated more accurately.
3) We can easily constrain that a charging node can only be visited by at most one EV, just like a customer node, so that the capacity constraint of a CS can be applied since a CS only has a limited number of charging piles for use.

III. Formulation of EVRP

We shall establish the optimization model for EVRP in this section based on graph $\mathcal{G}$.

A. Decision Variable and Objective Function

Since both customer requests and charging actions are modeled as nodes, EVRP essentially decides the best route,

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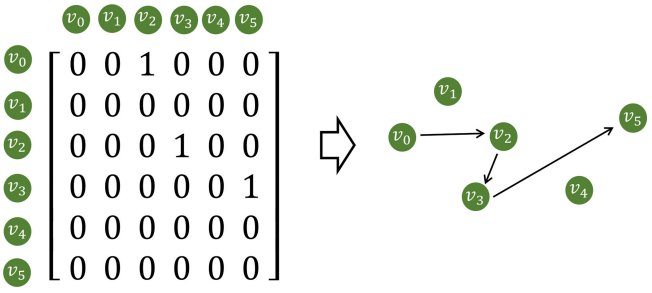


Fig. 2. Illustration of matrix $\mathbf{X}^k$.

along which an EV shall travel and visit a series of nodes, including both customer nodes and charging nodes, one by one. Let the binary decision variable x_{ij}^k denote whether EV k traverses the arc $a_{ij} \in \mathcal{A}$, as follows:

$$x_{ij}^k = \begin{cases} 1, \text{ if } EV\ k \text{ travels the arc } a_{ij} \\ 0, \text{ otherwise.} \end{cases} \tag{3}$$

Clearly, an $n \times n$ decision variable matrix $\mathbf{X}^k$ can summarize all the arc selections of EV k. For example, as shown in Fig. 2, only x_{02}^k, x_{23}^k and $x_{35}^k = 1$, which means that EV k starts from the start depot and visits sequentially the nodes v_2, v_3 and v_5 by traveling along the arcs a_{02}, a_{23} and a_{35}.

The objective function is shown in (4), which maximizes the profit of serving customers and operating the EV fleet

$$\mathcal{P}\left(x_{ij}^k\right) = \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{R}} x_{ij}^k m_j - \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{C}} x_{ij}^k w_j \tag{4}$$

where the revenue of visiting node j by EV k is denoted as $x_{ij}^k m_j$, thus the total revenue of serving all nodes by EV fleet is $\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{R}} x_{ij}^k m_j$. Similarly, $x_{ij}^k w_j$ is the cost of EV k charging at node j and $\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{C}} x_{ij}^k w_j$ denotes the total charging cost.

B. Network Flow Constraints and Battery Constraints

Since EVRP is investigated on the transportation network $\mathcal{G}$, a set of network flow constraints shall be satisfied as follows:

$$\sum_{j\in\mathcal{V}} x_{ij}^k - \sum_{j\in\mathcal{V}} x_{ji}^k = b_i \quad \forall i \in \mathcal{V}, k \in \mathcal{K},$$
$$b_0 = 1, b_n = -1, b_i = 0 \tag{5}$$

and

$$\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}} x_{ij}^k \le 1 \quad \forall j \in \mathcal{V}. \tag{6}$$

Constraint (5) states that each EV arrives at a customer node or a charging node, and must exit from the same node; therefore, $b_i = 0$. Since an EV only starts from the start depot and returns back to the end depot, we have $b_0 = 1$ and $b_n = -1$. At the same time, (6) constrains that a request/charging node can only be visited by one EV at most.

In addition, a set of battery constraints must also be considered in EVRP. Let E_j^k denotes the SOC of EV k at node j and it must satisfy the following box constraint:

$$E_j^k \in [0, E_{\max}] \quad \forall j \in \mathcal{V}, k \in \mathcal{K}. \tag{7}$$

Meanwhile, the dynamics of E_j^k along its path can be written as below according to most previous studies [22], [23]

$$E_j^k = \sum_{i\in\mathcal{V}}\Big(E_i^k + (g_i - p_{ij})\Big)x_{ij}^k \quad \forall j \in \mathcal{V}, k \in \mathcal{K} \tag{8}$$

where g_j denotes the charging amount of energy, and $p_{ij} = d_{ij}\epsilon$ denotes the energy consumed on arc a_{ij}.

In summary, the optimization model of EVRP can be formulated as Problem 1.

Problem 1 [Original EVRP (3-Index NIP)]:

$$\mathcal{P}^* = \max_{x_{ij}^k\in\{0,1\}} \mathcal{P}$$
$$\text{s.t.} \quad (5), (6), (7), (8). \tag{9}$$

Clearly, Problem 1 is an NIP and computationally challenging due to following reasons.

1) It is an integer programming problem.
2) Because the decision variable x_{ij}^k has three indices, the problem size is very large and at the level of KV^2, where K and V are the numbers of EVs and nodes.
3) The constraint (8) has many bilinear terms $(E_i^k + (g_i - p_{ij}))x_{ij}^k$, which further complicates the computation.

To this end, we propose a very efficient method to solve Problem 1 and yield a near-optimal solution in the next section.

IV. Solution Methodology

The key to solving Problem 1 efficiently lies in converting the original EVRP in NIP form to an approximate minimum cost flow problem in LP form. To this goal, this section presents a systematical approach to address computational challenges and eventually obtain a near-optimal solution. To be specific, we design a novel transformation procedure that contains three steps, as shown in Fig. 3.

1) First, we address the bilinear terms in constraint (8) by reformulating the SOC dynamics and applying Big-M technique, which lead to an equivalent linear IP problem.
2) Then, we relax the SOC box constraint (7) of the IP problem by introducing Lagrangian multipliers, and consider its approximate dual problem which only contains network flow constraints.
3) At last, we demonstrate that the three-index binary variable $x_{ij}^k \in \{0, 1\}$ can be equivalently converted into a two-index continuous variable $y_{ij} \in [0, 1]$, so that the problem eventually becomes an LP that enjoys very fast computation.

The above three steps of transformation will be detailed separately in Sections IV-A–IV-C.

The rationale behind the proposed method is to construct an LP problem as close to the original EVRP (9) as possible, so that both computational efficiency and high solution quality can be enjoyed. As shown in Fig. 3, after a set of transformations, the original NIP problem is eventually converted to an LP that even has much fewer decision variables, because each variable in Problem 4 has only 2 indices instead of 3 indices as in Problem 1. Since the original NIP now becomes an LP, and the number of decision variables is greatly reduced,

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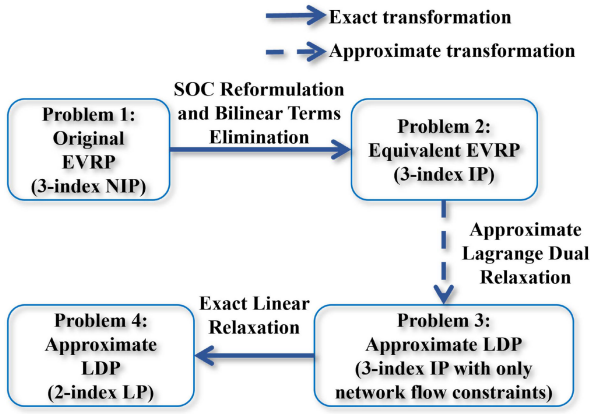


Fig. 3. Transformation procedure of proposed algorithm.

fast computational efficiency can be obtained. On the other hand, besides two exact transformations there is only one approximate transformation, which is approximate Lagrange dual relaxation. In general, this is often a very good approximation because it tries to minimize the approximation gap via iterations [24]. Therefore, the final optimization problem (Problem 4) approximates the original one very closely, so that the proposed method can preserve very good solution quality compared with other heuristic methods. Next, we shall present the transformation techniques, and analyze convergence and complexity of the proposed method.

A. Reformulating SOC Dynamics and Eliminating Bilinear Terms

As can be seen from in (8), SOC E_j^k contains a lot of bilinear terms $(E_i^k + (g_i - p_{ij}))x_{ij}^k$, which is inconvenient to handle and brings extra burdens to the computation. Therefore, we propose an equivalent yet novel reformulation of SOC dynamics as below, which can greatly reduce the number of bilinear terms.

Since the SOC change of EV k on arc a_{il} is $x_{il}^k(g_i - p_{il})$, the total SOC change from start depot to node j is $f_j^k \sum_{i=0}^{j}\sum_{l=0}^{j} x_{il}^k(g_i - p_{il})$, where $f_j^k \in \{0, 1\}$ indicates whether EV k visits node j or not. That is

$$f_j^k = \sum_{i\in\mathcal{V}} x_{ij}^k \quad \forall j \in \mathcal{V}, k \in \mathcal{K}. \tag{10}$$

For example and as shown in Fig. 4, $\sum_{i=0}^{j}\sum_{l=0}^{j} x_{il}^k(g_i - p_{il})$ accounts the upper diagonal submatrix of X^k, and the SOC change from start depot to node 3 in is $(g_2 - p_{02}) + (g_3 - p_{23})$.

Consequently, the SOC E_j^k can be rewritten as follows:

$$E_j^k = \left(\sum_{i=0}^{j}\sum_{l=0}^{j} x_{il}^k(g_i - p_{il}) + E_0\right) f_j^k = \delta_j^k f_j^k \tag{11}$$

where we define $\delta_j^k = \sum_{i=0}^{j}\sum_{l=0}^{j} x_{il}^k(g_i - p_{il}) + E_0$ for concise presentation. The above formulation simply accumulates the SOC change at all previous nodes, and does not rely on any historical knowledge of the route. More importantly, new formulation of E_j^k only contains one bilinear term, which is much less than that in traditional formulation (8).

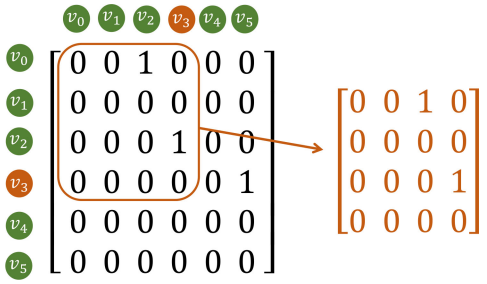


Fig. 4. SOC change on v_3.

With (11), the SOC box constraint (7) can be written as

$$0 \le \delta_j^k f_j^k \le E_{\max} \tag{12}$$

which still contains one bilinear term $\delta_j^k f_j^k$ that is an obstacle for fast computation. Then, the Big-M method is introduced to eliminate this bilinear term as follows:

$$\begin{cases} \delta_j^k - M\left(f_j^k - 1\right) \ge 0 \\ \delta_j^k + M\left(f_j^k - 1\right) \le E_{\max}. \end{cases} \tag{13}$$

The following analysis shows that (13) is equivalent to (12).

1) If $f_j^k = 0$, meaning EV k does not visit node j, both (12) and (13) impose no constraints on δ_j^k.
2) If $f_j^k = 1$, meaning EV k visits node l, both (12) and (13) require $0 \le \delta_j^k f_j^k \le E_{\max}$.

Thus, we obtain the following equivalent EVRP that is a linear IP problem without the bilinear term, as Problem 2. *Problem 2 [Equivalent EVRP (3-Index Linear IP)]:*

$$\begin{aligned} \mathcal{P}^* = \max_{x_{ij}^k \in \{0,1\}} \quad & \mathcal{P} \\ \text{s.t.} \quad & (5), (6), (13). \end{aligned} \tag{14}$$

B. Lagrange Relaxation and Approximate Dual Problem

In this step, we shall relax SOC constraint (13), so that only network flow constraints (5) and (6) are preserved. To this goal, Lagrangian multipliers are introduced for (13) and the corresponding Lagrangian function $\mathcal{L}$ is

$$\begin{aligned} \mathcal{L} = & \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}'}\sum_{j\in\mathcal{V}'} x_{ij}^k m_j - \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}'}\sum_{j\in\mathcal{C}'} x_{ij}^k w_j \\ & + \sum_{k\in\mathcal{K}}\sum_{j\in\mathcal{V}'} \alpha_j^k\left[\delta_j^k - M\left(f_j^k - 1\right)\right] \\ & + \sum_{k\in\mathcal{K}}\sum_{j\in\mathcal{V}'} \beta_j^k\left[E_{\max} - \delta_j^k - M\left(f_j^k - 1\right)\right] \end{aligned} \tag{15}$$

where $\boldsymbol{\alpha} \triangleq [\alpha_1^1, \ldots, \alpha_j^k]$ and $\boldsymbol{\beta} \triangleq [\beta_1^1, \ldots, \beta_j^k]$ are the Lagrangian multipliers for constraints $\delta_j^k - M(f_j^k - 1) \ge 0$ and $\delta_j^k + M(f_j^k - 1) \le E_{\max}$, respectively. Intuitively, the Lagrange multipliers α_j^k and β_j^k can be seen as penalties if the SOC of EV k at node j exceeds either the lower bound 0 or upper bound $E_{\max}$.

With $\mathcal{L}$, we can construct the Lagrange dual problem (LDP) of (14)

$$\mathcal{D}^* = \min_{\tilde{\alpha}_j, \tilde{\beta}_j} \max_{x_{ij}^k \in \{0,1\},\ (5),(6)} \mathcal{L}. \tag{16}$$

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According to the theorem of duality [24], $\mathcal{D}^*$ is a good and widely used approximate of $\mathcal{P}^*$, because $\mathcal{P}^*$ is always upper bounded by $\mathcal{D}^*$, i.e., $\mathcal{P}^* \leq \mathcal{D}^*$. Meanwhile, with proper selection of Lagrange multipliers α_j^k and β_j^k, the gap between $\mathcal{P}^*$ and $\mathcal{D}^*$ can be minimized.

However, the inner problem of (16) is still an IP problem with 3-index variable x_{ij}^k and difficult to solve. To this end, we make two modifications to $\mathcal{L}$. First, considering that any node can only be visited by one EV at most, we introduce an approximation of the multipliers as follows:

$$\alpha_j^k = \bar{\alpha}_j,\ \beta_j^k = \bar{\beta}_j. \tag{17}$$

Second, we introduce another penalty term $\bar{\gamma}_j \sum_{i\in\mathcal{V}} x_{ij}^k (g_i - p_{ij})$ that encourages an EV to have positive SOC increment from any node i to node j. As such and applying simple manipulations, the approximate Lagrangian function becomes

$$\begin{aligned}\mathcal{L}_A = &\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} x_{ij}^k m_j - \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{C}} x_{ij}^k w_j \\ &+ \sum_{j\in\mathcal{V}}(\bar{\alpha}_j + \bar{\beta}_j)\left[M\left(K - \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}} x_{ij}^k\right)\right] \\ &+ \sum_{j\in\mathcal{V}}(\bar{\alpha}_j - \bar{\beta}_j)\sum_{k\in\mathcal{K}}\left[\sum_{i=0}^{j}\sum_{l=0}^{j} x_{il}^k (g_i - p_{il}) + E_0\right] \\ &+ \sum_{j\in\mathcal{V}}\bar{\gamma}_j\left[\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}} x_{ij}^k (g_i - p_{ij})\right] \\ &+ K\sum_{j\in\mathcal{V}}\bar{\beta}_j E_{\max}.\end{aligned} \tag{18}$$

Above equation uses the fact $f_j^k = \sum_{i\in\mathcal{V}} x_{ij}^k$ from (10), and $K = |\mathcal{K}|$ represents the total number of EVs. Now the approximate LDP can be formulated as Problem 3.

Problem 3 [Approximate LDP (3-Index Linear IP With Only Network Flow Constraints)]:

$$\mathcal{D}_A^* = \min_{\bar{\alpha}_j,\bar{\beta}_j,\bar{\gamma}_j} \max_{x_{ij}^k\in\{0,1\},\ (5),(6)} \mathcal{L}_A. \tag{19}$$

Clearly, (19) is a two-level problem that requires iterations between the outer and inner subproblems. The outer subproblem is convex in $\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j$ according to duality theorem, so that proper values of $\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j$ can be solved via subgradient method as follows [24]:

$$\begin{cases}\bar{\alpha}_j(m+1) = \{\bar{\alpha}_j(m) + \xi_{\bar{\alpha}}(m)\nabla\mathcal{D}_A[\bar{\alpha}_j(m)]\}^+ \\ \bar{\beta}_j(m+1) = \{\bar{\beta}_j(m) + \xi_{\bar{\beta}}(m)\nabla\mathcal{D}_A[\bar{\beta}_j(m)]\}^+ \\ \bar{\gamma}_j(m+1) = \{\bar{\gamma}_j(m) + \xi_{\bar{\gamma}}(m)\nabla\mathcal{D}_A[\bar{\gamma}_j(m)]\}^+\end{cases} \tag{20}$$

where the iteration index and step sizes are denoted as m, $\xi_{\bar{\alpha}}(m)$, $\xi_{\bar{\beta}}(m)$ and $\xi_{\bar{\gamma}}(m)$, and the subgradients are denoted by $\nabla\mathcal{D}[\bar{\alpha}_j(m)]$, $\nabla\mathcal{D}[\bar{\beta}_j(m)]$ and $\nabla\mathcal{D}[\bar{\gamma}_j(m)]$, respectively.

On the other hand, we will also show next that the inner subproblem, i.e.,

$$\begin{aligned}\max_{x_{ij}^k\in\{0,1\}}\ &\mathcal{L}_A \\ \text{s.t.}\ &(5), (6)\end{aligned} \tag{21}$$

is equivalent to an LP and can be efficiently solved.

C. Equivalent Transformation to LP for Inner Subproblem

The advantage of using approximate multipliers is that for the inner subproblem (21), all coefficients including $\bar{\alpha}_j$, $\bar{\beta}_j$, $\bar{\gamma}_j$, m_j and w_j in both the objective function (18) and constraints (5) and (6) are irrelevant to k. In other words, all coefficients of (21) is independent to the selection of EV. With this property, we can drop index k from the inner subproblem and transform the three-index variable x_{ij}^k to an equivalent two-index variable y_{ij} [25]. Indeed, since a node j can only be visited by at most one EV, an arc a_{ij} can only be traversed by one EV at most. Therefore, we can denote

$$y_{ij} = \sum_{k\in\mathcal{K}} x_{ij}^k \quad \forall i,j \in \mathcal{V} \tag{22}$$

where the two-index variable y_{ij} indicates whether an arc a_{ij} has been traversed or not. Consequently, $\mathcal{L}_A$ in (18) can be rewritten with y_{ij} replacing $\sum_{k\in\mathcal{K}} x_{ij}^k$

$$\begin{aligned}\mathcal{L}_A^y = &\sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{V}} y_{ij} m_j - \sum_{i\in\mathcal{V}}\sum_{j\in\mathcal{C}} y_{ij} w_j \\ &+ \sum_{j\in\mathcal{V}}(\bar{\alpha}_j + \bar{\beta}_j)\left[M\left(K - \sum_{i\in\mathcal{V}} y_{ij}\right)\right] \\ &+ \sum_{j\in\mathcal{V}}(\bar{\alpha}_j - \bar{\beta}_j)\left[\sum_{i=0}^{j}\sum_{l=0}^{j} y_{il}(g_i - p_{il}) + KE_0\right] \\ &+ \sum_{j\in\mathcal{V}}\bar{\gamma}_j\left[\sum_{i=0}^{j} y_{ij}(g_i - p_{ij})\right] \\ &+ K\sum_{j\in\mathcal{V}}\bar{\beta}_j E_{\max}.\end{aligned} \tag{23}$$

Similarly, constraints (5) and (6) can be rewritten as

$$\begin{aligned}&\sum_{j\in\mathcal{V}} y_{ij} - \sum_{j\in\mathcal{V}} y_{ji} = b_i \quad \forall i \in \mathcal{V} \\ &b_0 = K, b_n = -K, b_i = 0\end{aligned} \tag{24}$$

and

$$\sum_{i\in\mathcal{V}} y_{ij} \leq 1 \quad \forall j \in \mathcal{V}. \tag{25}$$

It is noted in (24) that all EVs leave from the start depot and return to the end depot, thus $b_0 = K$ and $b_n = -K$.

As a result, the three-index inner subproblem is converted to an equivalent two-index subproblem as follows:

$$\begin{aligned}\max_{y_{ij}\in\{0,1\}}\ &\mathcal{L}_A^y \\ \text{s.t.}\ &(24), (25).\end{aligned} \tag{26}$$

Furthermore, (26) has the standard form of the minimum cost flow problem, thus it can be exactly relaxed to an LP problem as follows [22]:

$$\begin{aligned}\max_{y_{ij}\in[0,1]}\ &\mathcal{L}_A^y \\ \text{s.t.}\ &(24), (25).\end{aligned} \tag{27}$$

After transforming (21) equivalently to (27), the approximate LDP (19) is equal to Problem 4.

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Algorithm 1 (LLP Algorithm)

Initialize iteration index $m = 1$ and Lagrangian multipliers $\bar{\alpha}_j(m)$, $\bar{\beta}_j(m)$ and $\bar{\gamma}_j(m)$. Send $\bar{\alpha}_j(m)$, $\bar{\beta}_j(m)$ and $\bar{\gamma}_j(m)$ to inner problem. Solve inner problem (27) with $\bar{\alpha}_j(m)$, $\bar{\beta}_j(m)$ and $\bar{\gamma}_j(m)$ to obtain $y_{ij}(m)$. Update $\bar{\alpha}_j(m+1)$, $\bar{\beta}_j(m+1)$ and $\bar{\gamma}_j(m+1)$ according to (20). $y_{ij}(m-1)$ and $y_{ij}(m)$ are feasible solutions, and $\left|y_{ij}(m-1) - y_{ij}(m)\right| \leq \eta$, with η being the tolerance.

Problem 4 [Approximate LDP (2-Index LP)]:

$$\mathcal{D}_A^* = \min_{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j} \max_{y_{ij} \in [0,1],\ (24),(25)} \mathcal{L}_A^y. \tag{28}$$

As stated before, (28) can be solved very efficiently via iterations, because the outer subproblem is convex and can be computed according to (20) and the inner one is LP in nature. Then, the iterative computation process is summarized in Algorithm 1.

D. Convergence and Complexity Analysis

Next, we shall briefly analyze the convergence and complexity of using the above iterative algorithm to solve (28). First, rewrite (28) as

$$\min_{\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j} f\left(\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j\right) \tag{29}$$

where $f(\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j) = \max_{y_{ij} \in [0,1],(24),(25)} \mathcal{L}_A^y$ is a nonsmooth convex objective function in $\bar{\alpha}_j$, $\bar{\beta}_j$ and $\bar{\gamma}_j$. Since $f(\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j)$ is convex and all subgradients are bounded, it is widely known that $\bar{\alpha}_j(m)$, $\bar{\beta}_j(m)$, $\bar{\gamma}_j(m)$ converge to their optimum $\bar{\alpha}_j^*$, $\bar{\beta}_j^*$, $\bar{\gamma}_j^*$ under the updating law (20), if the step sizes $\xi_{\bar{\alpha}}(m)$, $\xi_{\bar{\beta}}(m)$ and $\xi_{\bar{\gamma}}(m)$ are properly selected and vanishing with m. For example, a widely used selection of step size is $\xi_{\bar{\alpha}}(m) = \xi_{\bar{\beta}}(m) = \xi_{\bar{\gamma}}(m) = (1/m)$, which produces a convergence rate of $\mathcal{O}(m^{-[1/2]})$ [26]. Therefore, the algorithm can converge to a near-optimal solution in a finite number of steps.

As for the time complexity of Algorithm 1, we shall consider both the time complexity of each iteration and the number of iterations to reach a certain tolerance level. In each iteration, the main task is to solve the inner LP problem (27), whose time complexity is $\mathcal{O}(N^{3.5})$ using the interior method with N being the number of decision variables [27]. Since the decision variable is y_{ij}, we have $N = V^2$ where V is the number of nodes. As for the number of iterations, it is of the order of $\mathcal{O}([1/\eta^2])$ since $f(\bar{\alpha}_j, \bar{\beta}_j, \bar{\gamma}_j)$ is convex, where η is the tolerance [26], [28]. In summary, the total time complexity of Algorithm 1 is $\mathcal{O}([V^7/\eta^2])$.

In each iteration of Algorithm 1, computer memories are needed to store variables, parameters, and constraints. To this end, the complexity of the space is also briefly analyzed here. We shall first consider the memory needed for updating the multipliers $\bar{\alpha}_j(m)$, $\bar{\beta}_j(m)$, $\bar{\gamma}_j(m)$ according to (20). Since we have V nodes in total, the number of multipliers is $3V$ and the memory needed for (20) is $\mathcal{O}(V)$. On the other hand, solving LP problem (27), the total memory needed is $\mathcal{O}(CN + (C + N)^2)$, where C is the number of constraints [27]. Clearly, there are $2V$ network flow and task constraints (24) and (25), and $2N = 2V^2$ lower and upper bounds constrains for decision variables y_{ij}. Therefore, $C = \mathcal{O}(V^2)$ and the space complexity of Algorithm 1 is $\mathcal{O}(V^4)$.

In summary, Algorithm 1 is convergent and can reach to a near-optimal solution in a finite number of steps. Meanwhile, both its time complexity and space complexity are polynomials. Therefore, it is more suitable to solve large-scale problems than conventional methods as to be illustrated next. Meanwhile, it is worth noting that the complexities are irrelevant to the number of EVs because all EVs are homogeneous and we use y_{ij} to equivalently replace $\sum_{k \in \mathcal{K}} x_{ij}^k$ as the variable so that the final optimization problem (28) does not contain EV index k.

V. Numerical Simulation

In this section, we evaluate the efficacy of the proposed algorithm across 14 instances, which differ in terms of the number of EVs, the number of customers, the number of charging piles, and the length of time slots, as shown in Tables II and III. The experiments are conducted using four different methods.

1) Lagrangian LP (LLP) algorithm proposed in this article.
2) ALNS, which is a heuristic algorithm known for its swift problem-solving capabilities.
3) Branch-and-Price-and-Cut (BPC) algorithm, which is a widely used analytical approach.
4) NIP, where we directly solve the original EVRP using the commercial solver Gurobi.

Notably, both BPC and NIP are exact algorithms and capable of achieving global optimal solutions to serve as benchmark for comparing optimality; however, they often suffer from heavy computation burdens. On the other hand, ALNS is faster than BPC and NIP, at the cost of losing a large amount of optimality sometimes. In contrast, the proposed LLP is even faster than ALNS, and the optimality is well preserved. All methods are implemented with Python 3.7 on a PC with Intel Core i5-7500 3.40 GHz and 12 GB of RAM.

A. Parameter Settings

To showcase the simulation results, we first present the parameter settings. We adopt the map data of Belgium [29], and then insert different numbers of customer requests, EVs, and charging piles randomly into the map. The visiting times of customer requests are also generated randomly and sent to the EV dispatcher before EVs begin service.

As for EVs, we assume the battery capacity of an EV is $E_{\max} = 40$ kWh, and the initial energy of all EVs is full. All EVs have the same energy consumption rate $\varepsilon = 0.159$ kWh/km [30], and the same velocity $v^k = 100$ km/h $\quad \forall k \in \mathcal{K}$. The amount of charging during a one-time slot is $g_j = 24$ kWh. The time horizon for EVRP is set from 8:00 to 18:00. We set the income to be 20€ for serving a customer, and the electricity prices will fluctuate hourly in accordance with Belgium market rates [31] as illustrated in Fig. 5. The maximum computation time is set to a 7200-s time limit, and

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TABLE II
SIMULATION DATA OF THE COMPARISON EXPERIMENT OF 30 MIN TIME SLOT

Instance	Time slot (min)	# of EVs	# of customers	# of CSs	# of charging piles in total	**# of nodes**
1	30	2	5	1	3	67
2	30	2	10	1	3	72
3	30	3	20	1	3	82
4	30	3	30	1	3	92
5	30	10	50	1	3	112
6	30	10	100	2	6	222
7	30	20	200	3	9	382

TABLE III
SIMULATION DATA OF THE COMPARISON EXPERIMENT OF 15 MIN TIME SLOT

Instance	Time slot (min)	# of EVs	# of customers	# of CSs	# of charging piles in total	**# of nodes**
8	15	2	5	1	3	127
9	15	2	10	1	3	132
10	15	3	20	1	3	142
11	15	3	30	1	3	152
12	15	10	50	1	3	172
13	15	10	100	2	6	342
14	15	20	200	3	9	562

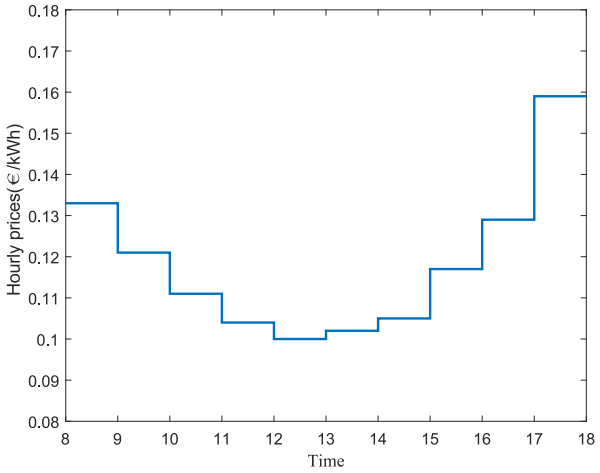


Fig. 5. Dynamic electricity prices in Belgium from 8:00 to 18:00.

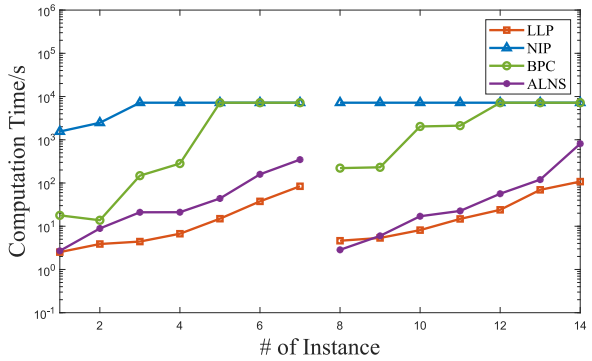


Fig. 6. Comparison on computation time. In most instances, the proposed LLP is about 100 times faster than BPC and several times faster than ALNS.

it is seen from later results that NIP and BPCP cannot even solve some instances within the time limit.

B. Simulation Results

For the sake of clarity in comparison, our primary evaluation criteria include computation time, accrued profit and the number of served customers and the results are detailed next.

1) Computation Time: Fig. 6 compares the computation time of LLP, BPC, NIP, and ALNS. The results indicate that NIP is very slow and the computation time for most cases exceed the 7200-s time limit. BPC is a bit faster than NIP, but its computation time increases rapidly with the number of nodes and exceeds the time limit when the problem has more than 50 customers. This observation can be attributed to the escalating complexity of the problem with the expansion of the transportation network. In contrast, our proposed algorithm is about 100 times faster than BPC for most instances, and also several times faster than the widely used ALNS. The computation time of our method is less than 100 s even for instances 7 and 14 that have 200 customer requests and 20 EVs.

2) Profit: As shown in Fig. 6, the exact algorithms such as BPC and NIP can only find the optimal solutions within the time limit for small-scale instance, i.e., instances 1–4 and 8–11, therefore we present in Fig. 7 the profit obtained by BPC (the same as NIP), LLP and ALNS for small-scale instances. It is seen that the optimality gap between LLP and BPC is very small, while ALNS yields less profit. The relative optimality gap of LLP robustly remains below 7%, but that of ALNS can reach as high as 40%.

As for other large-scale instances, i.e., instances 5 to 7 and 12 to 14, we can only compare the profit obtained by LLP and ALNS. As shown in Fig. 8, LLP outperforms ALNS in all large-scale instances, which suggests that LLP is more suitable to EVRP than the widely used ALNS.

3) Number of Served Customers: The number of served customers is also an important criterion for evaluating different algorithms. Similar to the previous analysis, BPC is only able to obtain optimal solutions for small-scale cases, and the proposed LLP can serve a similar number of customers, as shown in Fig. 9. In contrast, ALNS only serves fewer

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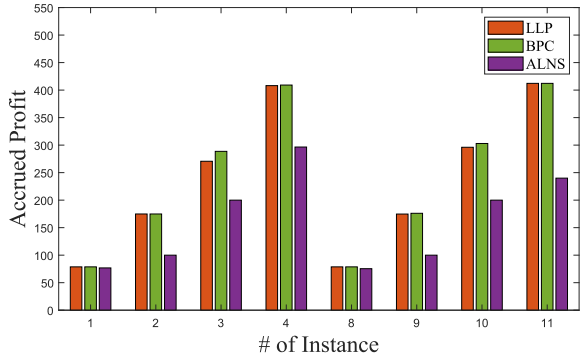


Fig. 7. Comparison on profit for small-scale instances. The proposed LLP is close to the optimal value of BPC, and much better than ALNS.

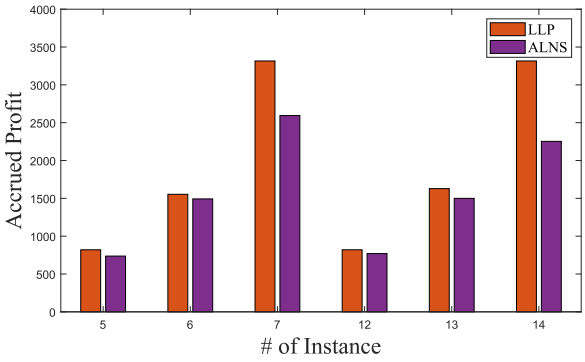


Fig. 8. Comparison on profit for large-scale instances. The proposed LLP is better than ALNS.

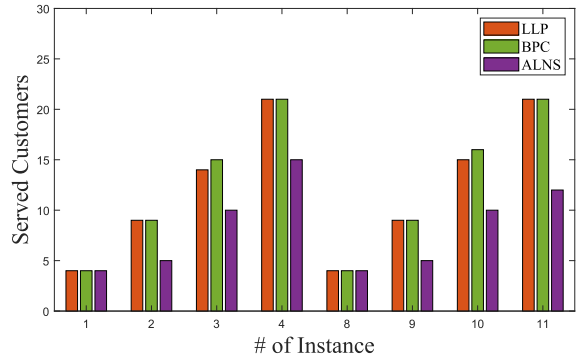


Fig. 9. Comparison on served customers for small-scale instances. The proposed LLP is close to the optimal value of BPC, and much better than ALNS.

customers. Fig. 10 compares LLP and ALNS for large-scale instances, and it shows that LLP is better than ALNS.

In summary, the simulation results suggest that the proposed LLP is capable of solving EVRP with over 500 nodes within 100 s, and is much faster than BPC but still preserves very good optimality. Meanwhile, LLP also outperforms the widely used ALNS in both computation time and optimality.

VI. Conclusion

In this article, we investigate EVRP while considering CS capacity and dynamic electricity price in particular. After establishing a new graph model that also treats the charging actions as nodes and charging prices as costs on the nodes,

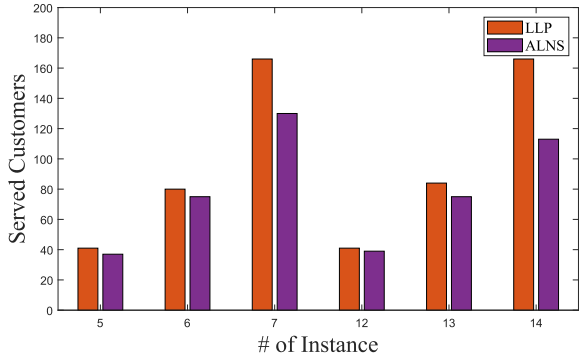


Fig. 10. Comparison on served customers for large-scale instances. The proposed LLP is better than ALNS.

we formulate EVRP as a three-index binary NIP problem on a directed graph. However, solving the problem directly is computationally intensive. To obtain an efficient solution, a novel algorithm consisting a series of transformations is developed to transform the original problem into a computationally efficient LP problem. The proposed algorithm is evaluated through comparison simulations, and the results demonstrate its good performance in terms of computation speed, profit, and the number of served customers. Because of its fast computational speed, this method is particularly suitable for rolling horizon framework in practical applications where EVRP needs to be solved frequently.

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